

1. Overview for AudioPeakRMSChecker.ps1 v5.1.1

This script analyzes the audio tracks inside a movie file (MKV, MP4). It checks for:

Peak (dB) loudest moment in the audio.

RMS (dB) average loudness over time.

Crest Factor (dB) difference between Peak and RMS

Clipping (number of samples that hit or exceed 0 dB)

ebur128 — EBU R128 loudness, reported only, never gates:

- Integrated loudness (I) in LUFS, with its gating threshold.
- Loudness Range (LRA) in LU, with range threshold and LRA low/high.
- True Peak in dBTP.

These loudness values are informational only: they never affect the PASS/FAIL result or the exit code.

The function is to give you a quick PASS/FAIL result so you can tell whether the audio is clean, safe, and ready for encoding, downmixing, or playback. It supports common audio formats, including TrueHD (7.1 / 5.1), DTS-HD MA (7.1 / 5.1), AAC (7.1 / 5.1 / 2.0), Dolby Digital Plus (7.1 / 5.1 / 2.0), Dolby Digital (7.1 / 5.1 / 2.0), and Opus 2.0.

This tool does **not** modify audio. It only analyzes it and reports the results.

2. Requirements

PowerShell 7.6 - [Click here to Download PS 7.6 \(Windows x64 .msi\)](#)

FFmpeg 8.1 - <https://ffmpeg.org/download.html>

Quick Start (Windows 11)

1. Place **AudioPeakRMSChecker.ps1**, **ffmpeg.exe**, and **ffprobe.exe** in the same folder.
2. Run the script:

```
pwsh -ExecutionPolicy Bypass -File .\AudioPeakRMSChecker.ps1 ".\YourMovie.mkv"
```

The -ExecutionPolicy Bypass only applies to this single run and doesn't change your system settings.

macOS / Linux

1. Place the script, **ffmpeg**, and **ffprobe** in the same folder.
2. Make the binaries executable (first time only):

```
chmod +x ffmpeg ffprobe
```

3. Run:

```
pwsh -File ./AudioPeakRMSChecker.ps1 "./YourMovie.mkv"
```

3. Supported Audio Formats

AudioPeakRMSChecker analyzes the following audio formats:

Codec Family	Formats Included
TrueHD	TrueHD 7.1, TrueHD 5.1
AAC	AAC 7.1, AAC 5.1, AAC 2.0
EAC3 / DDP	Dolby Digital Plus 7.1, Dolby Digital Plus 5.1, Dolby Digital Plus 2.0
AC3 / DD	Dolby Digital 7.1, Dolby Digital 5.1, Dolby Digital 2.0
DTS	DTS-HD MA 7.1, DTS-HD MA 5.1
Opus	Opus 2.0

Note on DTS: The script detects DTS tracks by codec name and channel count. It cannot confirm at the probe level whether a DTS track is DTS Core or DTS-HD MA — both are reported as `dts` by `ffprobe`. The display label reads "DTS-HD MA" by channel count as a naming convention; it does not confirm the track is DTS-HD MA rather than DTS Core.

The script automatically detects:

- the codec
- the number of channels
- whether the track is a source (7.1) or downmix (5.1)
- whether the track is supported for analysis
- Unsupported tracks are ignored safely.

AAC 5.1 selection rule: When a file has both a 7.1 source track and a DDP 5.1 (or DD 5.1) downmix track, AAC 5.1 is skipped — those two tracks already cover source and downmix analysis. In all other cases, AAC 5.1 is analyzed normally.

4. How the Analysis Works

The script performs the following steps:

1. Scans the movie file

It looks for audio tracks and identifies their codec and channel layout.

2. Runs FFmpeg's "astats" and "ebur128" filters

In a single decode pass per track, **astats** measures peaks, RMS, clipping, and dynamic range (the PASS/FAIL gate), and **ebur128** measures EBU R128 loudness (reported only).

3. Collects the results

The script extracts the important numbers from FFmpeg's output.

4. Checks for problems

It looks for:

- peaks too close to 0 dB
- audio that is too loud overall
- low dynamic range
- clipped samples

5. Returns PASS or FAIL

PASS means the audio is clean and safe.

FAIL means the audio may distort or has loudness issues.

6. Saves a log file

A text file is created with the **astats** output and the EBU R128 Summary for reference (the high-frequency ebur128 per-frame lines are stripped).

"Movie-source-**astats.txt**", "Movie-downmix-**astats.txt**" depending on which track was analyzed.

Single-track runs use "Movie-single-**astats.txt**."

A manual `-Track N` run uses "Movie-trackN-**astats.txt**".

5. What Each Metric Means

AudioPeakRMSChecker reports four main metrics:

Peak (dB)

- The loudest moment in the audio.
- Values **near 0 dB** mean the audio is very loud.
- Values **above 0 dB** indicate clipping.

RMS (dB)

- The average loudness over time.
- Higher RMS (closer to 0 dB) means the audio is louder.
- Lower RMS (–20 dB to –35 dB) means more dynamic range.

Crest Factor (dB)

- The difference between Peak and RMS.
- Higher crest = more dynamic, cinematic sound.
- Low crest = compressed or flat audio.

Clipped Samples

- The number of samples that hit or exceed 0 dB.
- A few clips are usually harmless.
- Many clips indicate distortion.

These metrics help determine whether the audio is clean or risky.

The following three metrics come from the ebur128 filter. They are **reported only and never gate** PASS/FAIL:

Integrated Loudness (I, LUFS)

- Program loudness averaged across the whole track using EBU R128 gating.
- Lower (more negative) means a quieter overall program.
- Reference targets: EBU R128 –23 LUFS, ATSC A/85 –24 LUFS, streaming –14 to –16 LUFS.

Loudness Range (LRA, LU)

- The statistical spread between the quiet and loud passages of the program.
- Higher means more dynamic; lower means more compressed.

True Peak (dBTP)

- Inter-sample peak estimate; it can read higher than the sample Peak above.
- Values above 0 dBTP risk clipping on D/A conversion or on a lossy re-encode.

5.5 Quick Interpretation

Each track gets a **Quick Interpretation** that puts the numbers into plain language.

Peak ranges:

- 0 dB and above — at or above 0 dB (risk of clipping) [WARNING]
- -1 to 0 dB — very hot (close to clipping) [WARNING]
- -3 to -1 dB — healthy (good headroom) [PASS]
- Below -3 dB — low (very safe headroom) [PASS]

RMS ranges:

- Above -12 dB — very loud (compressed mix) [WARNING]
- -20 to -12 dB — moderately loud (typical TV mix) [PASS]
- -30 to -20 dB — normal for movies (healthy dynamics) [PASS]
- Below -30 dB — very low (quiet or highly dynamic) [PASS]

Crest factor ranges:

- Below 6 dB — heavily compressed [WARNING]
- 6 to 12 dB — balanced dynamics [PASS]
- 12 to 20 dB — good dynamic range [PASS]
- Above 20 dB — extremely dynamic (very wide range) [PASS]

Clipping interpretation:

- More than 50 clips at ≥ 0 dB — audio distortion likely [WARNING]
- 1 or more clips at ≥ 0 dB — clipped samples detected [WARNING]
- 6 or more clips below 0 dB — minor distortion possible [WARNING]
- 1 to 5 clips — not clipping [PASS]
- 0 clips — clean [PASS]

For TrueHD tracks, the Quick Interpretation also shows the preroll offset in milliseconds. This is a normal PTS offset, not a sync issue.

TrueHD Preroll and A/V Sync

TrueHD audio tracks often report a **negative start_time** (for example, -0.078 seconds). This is normal behavior for the TrueHD format and does **not** indicate an A/V sync issue.

AudioPeakRMSChecker.ps1 converts this value to milliseconds.

Preroll: -78 ms — normal TrueHD PTS offset (source); not a sync issue

Negative preroll simply means the TrueHD stream begins slightly before the video so the decoder can initialize properly.

A/V Sync Tolerances for TrueHD:

- Up to **120 ms before** the video → **safe / expected preroll**
- Up to **5 ms after** the video → **safe (container rounding)**
- Outside this range → **A/V Sync WARNING**

Negative preroll values within normal range are expected and safe. They do **not** require correction and do **not** indicate misalignment.

6. PASS / FAIL Rules (Simple)

The script marks a track as **FAIL** if any of the following are true:

- **Peak > 0 dB**
The audio is clipping.
- **RMS louder than -12 dB**
The audio is extremely loud and may distort.
- **Crest < 6 dB**
The audio is heavily compressed.
- **Clipped samples > 5**
Too many clipped peaks.
- **Peak $\geq$ 0 dB with any clipped samples**
Even a single clipped sample triggers FAIL when the peak is at or above 0 dB.

If none of these conditions are met, the track is marked **PASS**.

PASS means:

- clean audio

- safe headroom
- healthy dynamics
- no distortion

FAIL means:

- the audio may distort
- the mix may be too loud
- the track may need correction

7. JSON Output Mode

Useful for: Debugging and troubleshooting (easier to inspect raw values).

In JSON mode: No summary is printed, No color output, No interpretation text.

```
$DefaultJsonMode = $false (top of the script)
To **enable** JSON mode by default: change `$false` to `$true` on line
70
To **disable** JSON mode by default: change `$true` back to `$false` on
line 70
```

You can also enable JSON mode for a single run without editing the script:

```
pwsh -File .\AudioPeakRMSChecker.ps1 ".\YourMovie.mkv" -JsonMode
```

8. Source vs Downmix Comparison

When a file has both a 7.1 source track and a DDP 5.1 (or DD 5.1) downmix, the script prints a side-by-side comparison after the individual results:

```
=== SOURCE vs DOWNMIX (peak / RMS) ===
Source Peak: -1.2 dB
Downmix Peak: -0.8 dB
Source RMS: -24.3 dB
Downmix RMS: -23.1 dB
Source Crest: 23.1 dB
Downmix Crest: 22.3 dB
```

This makes it easy to confirm the downmix is consistent with the source and hasn't introduced loudness or clipping issues.

When both tracks also carry ebur128 loudness, a second comparison follows. It is informational only and never gates PASS/FAIL:

```
=== SOURCE vs DOWNMIX LOUDNESS ===
```

	SOURCE	DOWNMIX
Integrated (I):	-24.3 LUFS	-23.8 LUFS
Loudness range (LRA):	12.1 LU	11.4 LU

```
delta integrated: +0.5 LU
```

```
delta LRA: -0.7 LU
```

```
tolerance: 1.0 LU
```

```
VERDICT: PRESERVED (|delta I| <= tolerance)
```

The deltas are `downmix - source`, so a positive integrated delta means the downmix is louder. The **verdict** reads `PRESERVED` when the absolute integrated-loudness delta is within tolerance, otherwise `SHIFTED`. The tolerance window defaults to 1.0 LU and is adjustable with `-Tolerance`:

```
pwsh -File .\AudioPeakRMSChecker.ps1 ".\YourMovie.mkv" -Tolerance 1.5
```

An interpretation block follows the verdict, restating the reference targets (EBU R128 -23 LUFS, ATSC A/85 -24 LUFS, streaming -14 to -16 LUFS), whether the downmix is louder or quieter than the source, and whether its loudness range is wider or narrower.

9. A/V Sync Check

After analysis, the script checks whether the audio starts in sync with the video and prints the result:

```
---A/V SYNC CHECK---  
A/V Sync : OK  
→ Safe to proceed with ConvertAudioEngine-DDP51 or ConvertAudioEngine-Keep71.
```

If sync is outside tolerance, it prints a warning instead:

```
A/V Sync : WARNING — audio starts +180 ms after video  
→ Do NOT run ConvertAudioEngine-DDP51 or ConvertAudioEngine-Keep71 until this  
sync issue is fixed.
```

Tolerances used:

Track type	Allowed range	Reason
TrueHD	Up to 120 ms before video	Normal decoder preroll
TrueHD	Up to 5 ms after video	Container rounding
All others	Within ± 5 ms	Standard tolerance

The sync check uses the primary 7.1 source track when present, or falls back to the first audio stream. If the file has no video stream or timing data is unavailable, the sync check is skipped silently.

10. Tunable timeout parameters

\$MaxRuntimeMultiplier = 4, Multiplier applied to the ETA to set the FFmpeg timeout.
(user adjustable from 2 to 8)

\$MinTimeoutSeconds = 60, Minimum timeout floor per track, regardless of ETA.
(user adjustable from 30 to 300 seconds)

11. Exit Codes

number	description
0	Success
1	Input validation: file missing / no audio streams / no supported codecs / no valid 7.1 or 5.1, or 2.0 combination
2	FFmpeg produced no stderr output (analysis aborted)
3	FFmpeg exited with non-zero exit code
5	FFmpeg exceeded maximum allowed runtime (timeout guardrail)
6	astats output missing required metrics (Peak / RMS / Peak count)
7	Peak level dB failed numeric parse
8	RMS level dB failed numeric parse
9	Peak count failed numeric parse
10	astats log file was not created on disk
11	astats log file is empty or smaller than 10 bytes
12	ffmpeg or ffprobe binary missing or not executable in the script folder

Disclaimer

- Always test on a few files before batch-processing your library.
- Keep backups of original media.
- Results may vary depending on source quality and hardware.